

Lecture notes: Lang's FCC and Jacobians

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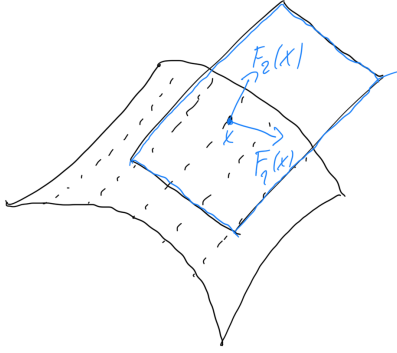
Abstract

We will go over the basic ideas of the paper [3], mainly concerning the passage from the "distributional" problem concerning the metric currents to the "functional" one concerning Jacobians of Lipschitz functions. Slightly less detail will be provided for showing the lack of regularity for the Jacobian problem.

1 Currents and Jacobians

1.1 Intro to Lang's currents

Goal: Encode information about a surface into only metric structure of $\mathbb{R}^d$.



$$\langle M \rangle(f, \pi^1, \pi^2) = \int_M f \det\left(\frac{\partial \pi^j}{\partial F_i}\right)_{i,j=1}^2 d\mathcal{H}^2$$

Makes sense for Lipschitz scalars f, π^1, π^2 by Rademacher!

$$(L) \quad \pi^1 \text{ or } \pi^2 \text{ const. on spt } f \implies \langle M \rangle(f, \pi^1, \pi^2) = 0$$

$$(JC) \quad f_i, \pi_i^1, \pi_i^2 \rightarrow f, \pi^1, \pi^2 \text{ pointwise with uniformly bounded Lipschitz constants} \implies \langle M \rangle(f_i, \pi_i^1, \pi_i^2) \rightarrow \langle M \rangle(f, \pi^1, \pi^2)$$

Definition 1.1 (Metric currents). Suppose X is a compact metric space. A metric k -current on X is a multilinear map

$$T: \text{Lip}(X) \times \text{Lip}(X)^k \rightarrow \mathbb{R} \quad (1.1)$$

(elements of $\text{Lip}(X)^k$ are denoted by $\pi = (\pi^1, \dots, \pi^k)$) satisfying

(L) If there is some $j \in \{1, \dots, k\}$ such that $\pi_j = 0$ on a neighbourhood of $\text{spt } f$, then $T(f, \pi) = 0$.

(JC) If each π_i^j converges to π^j and f_i converges to f (pointwise with bounded Lipschitz constants), then

$$T(f_i, \pi_i) \rightarrow T(f, \pi). \quad (1.2)$$

Space of all metric k -currents on X is denoted by $\mathcal{D}_d = \mathcal{D}_d(X)$.

Remark 1.2. For compact metric spaces the definition provided is equivalent to that of Lang [2]. To recover the definition of Ambrosio and Kirchheim [1], one needs to add the axiom of finite mass, which we will not discuss in detail. If X is merely locally compact or even just complete, the definition may be adjusted to yield similar theory in this more general setting.

Conjecture 1.3. Every metric k -current in $\mathbb{R}^d$ supported inside a compact set is a flat k -chain in the sense of Federer-Fleming.

Simplification in the top dimension Let $Q = [0, 1]^d \subset \mathbb{R}^d$ and suppose $u \in L^1(Q)$. We define

$$\text{Em}_*(u): \text{Lip}(Q) \times \text{Lip}(Q)^d \rightarrow \mathbb{R} \quad (1.3)$$

by

$$\text{Em}_*(u)(f, \pi) = \int_Q u f \det D\pi d\mathcal{L}^d. \quad (1.4)$$

This *always* defines a metric d -current.

Conjecture 1.4 (Lang's conjecture for top currents). *For every metric d -current T in Q there is some $u \in L^1(Q)$ such that $T = \text{Em}_* u$. In other words, $\text{Em}_*: L^1 \rightarrow \mathcal{D}_d$ is surjective.*

Now we will try to understand more about the operator Em_* by studying the dual operator. To do that we need to develop a theory of “potential forms” - objects on which metric currents act.

1.2 Metric compactum, normability of currents

Preliminaries Let X be a metric space with a distinguished point $0 \in X$.

$$\text{Lip}_0(X) = \{f: X \rightarrow \mathbb{R} \text{ Lipschitz}, f(0) = 0\} \quad (1.5)$$

is normed with $\text{Lip}(\cdot)$.

$$\text{Lip}_b(X) = \{f: X \rightarrow \mathbb{R} \text{ Lipschitz and bounded}\} \quad (1.6)$$

is normed with $\|\cdot\|_L := \max\{\text{Lip}(\cdot), \|\cdot\|_\infty\}$. Note that if X is compact then $\text{Lip}_b(X)$ coincides with the space of all Lipschitz maps $\text{Lip}(X)$.

Proposition 1.5. *Both spaces above are dual Banach spaces. If X is separable, then the preduals are separable. The weak* convergence on both of the unit balls is pointwise convergence. If X is compact, then the weak* convergence on both of the balls is metrized by*

$$d(f, g) = \|f - g\|_\infty, \quad (1.7)$$

so in particular, the unit balls equipped with this metric are compact (cf. Arzela-Ascoli).

Metric compactum From now on, X is a compact metric space.

Definition 1.6 (Metric compactum). For a compact metric space X we define the *metric k -compactum*

$$\mathcal{K}_k = (B_{\text{Lip}_b(X)}, w^*) \times (B_{\text{Lip}_0(X)}, w^*)^k \quad (1.8)$$

equipped with the (compact) product topology.

Note that by (JC) every metric k -current on X is an element of $C(\mathcal{K}_k)$.

Norm on the space of metric k -currents We let

$$\|T\|_{\mathbb{P}_d} = \|T\|_{\mathbb{P}_d(X)} = \|T\|_{C(\mathcal{K}_k)} \quad (\text{the latter being the supremum norm})$$

The norm is called the *potential norm*, currents for which it is finite are called potential currents. If X is compact, all metric currents are potential. The space of all potential currents is denoted by $\mathbb{P}_k$.

Remark 1.7. If X is only locally compact, similar theory makes sense but not every metric current is potential. Potential currents are those metric currents that “do not explode at the boundary/infinity”. This justifies renaming metric to potential in this setting.

Theorem 1.8 (Defining property of potential norm). *Suppose T is a metric current and $(f, \pi) \in \text{Lip}_b(X) \times \text{Lip}_0(X)^k$. Then*

$$|T(f, \pi)| \leq \|T\|_{\mathbb{P}_k} \max\{\text{Lip}(f), \|f\|_\infty\} \prod_{j=1}^k \text{Lip}(\pi^j). \quad (1.9)$$

Proof. If $(f, \pi) \in \mathcal{K}_k$, then it follows by definition. The general case follows from multilinearity, in particular the homogeneity in each variable. $\square$

Lemma 1.9. *The space $\mathbb{P}_k$ is complete.*

Proof. It suffices to show that it is a closed subspace of $C(\mathcal{K}_k)$. Suppose T_i is a sequence of metric k -currents converging to T in $C(\mathcal{K}_k)$. By definition, the limit is then jointly continuous. Checking that it is local and multilinear is a near trivial exercise. $\square$

1.3 Potential forms

We now specialize to $X = Q = [0, 1]^d$ and to d -currents and d -forms.

Formal Lipschitz sums A *formal Lipschitz sum* is an expression of the form

$$\sum_i (f_i, \pi_i), \quad (1.10)$$

where each f_i is a Lipschitz map on Q , each $\pi_i = (\pi_i^1, \dots, \pi_i^d)$ is Lipschitz with $\pi_i(0) = 0$ and such that

$$s(\sum_i (f_i, \pi_i)) = \sum_i \max\{\text{Lip}(f_i), \|f_i\|_\infty\} \prod_{j=1}^d \text{Lip}(\pi_i^j) < \infty \quad (1.11)$$

The space of all formal Lipschitz sums (with naturally defined addition and scalar multiplication, i.e. a vector space) is denoted by $\mathcal{A}_d$. Next we need to identify formal Lipschitz sums defining the “same differential form”. The equivalence relation is given by $\sum_i (f_i, \pi_i) \sim \sum_i (g_i, \kappa_i)$ if

$$\sum_i T(f_i, \pi_i) = \sum_i T(g_i, \kappa_i) \quad \text{for every } T \in \mathcal{D}_d = \mathbb{P}_d.$$

Note that due to Theorem 1.8 the sums converge absolutely.

Potential differential forms

Definition 1.10. We define the *potential norm* on $\mathcal{A}_d / \sim$ via

$$\|\omega\|_{\mathbb{P}^d} = \|\omega\|_{\mathbb{P}^d(Q)} = \inf\{s(\sum_i (f_i, \pi_i)) : \sum_i (f_i, \pi_i) \in \omega\} \quad \text{for } \omega \in \mathcal{A}_d / \sim.$$

The set $\mathbb{P}^d(Q) = \mathbb{P}^d = \mathcal{A}_d / \sim$, equipped with the potential norm, is called *the space of (top dimensional) potential (differential) forms*¹. Its elements are called *(top dimensional) potential (differential) forms*. The map

$$\text{Em}: \mathbb{P}^d(Q) \rightarrow L^\infty(Q),$$

given by $\text{Em}([\sum_i (f_i, \pi_i)]) = \sum_i f_i \det D\pi_i$ is called *the canonical embedding (for potential forms)*.

Remark 1.11. Somewhat confusingly, we have not actually proved that the canonical embedding is an embedding. I do not know how to show this, so it is left open (I will still call it an embedding though). The point is to prove that if one has two formal Lipschitz sums $\sum_i (f_i, \pi_i)$ and $\sum_i (g_i, \kappa_i)$ such that

$$\sum_i f_i \det D\pi_i = \sum_i g_i \det D\kappa_i, \quad (1.12)$$

then it also holds that $\sum_i (f_i, \pi_i) \sim \sum_i (g_i, \kappa_i)$. This would be true, for example, if $\text{Em}_*(L^1)$ were dense in $\mathbb{P}_d$.

Note that there is the natural *duality pairing* of potential currents and potential forms:

$$\langle T, \omega \rangle = \sum_i T(f_i, \pi_i) \quad \text{where } \sum_i (f_i, \pi_i) \in \omega. \quad (1.13)$$

Conjecture 1.12 (“Metric Wolffe’s theorem”). $\mathbb{P}^d = \mathbb{P}_d^*$. *Formally, to every element $\Phi \in \mathbb{P}_d^*$ there exists a potential form $\omega \in \mathbb{P}_d$ such that*

$$\langle T, \omega \rangle = \Phi(T) \quad \text{for every } T \in \mathbb{P}_d. \quad (1.14)$$

¹If one is familiar with the basic theory of tensor products, it can be easily verified, that the space $\mathbb{P}^d$ may alternatively be obtained from the tensor product, say $\mathcal{T} = \text{Lip}_b(Q) \otimes \bigotimes_{j=1}^d \text{Lip}(Q)$. This is due to the fact that the equivalence relation we define has strictly larger equivalence classes than the equivalence relation used to define the tensor product. Thus $\mathbb{P}^d = \overline{\mathcal{T}} / \sim$

If the conjecture were true, we would be almost done with the “abstract” nonsense part. Since we cannot prove it we need to develop some more theory to actually pass to the dual problem concerning Jacobians. What we are basically trying to do is pass to the dual problem *without* identifying the dual space. This is possible but involved.

Let $I: \mathbb{P}^d \rightarrow \mathbb{P}_d^*$ be given by $I(\omega)(T) = \langle T, \omega \rangle$. Then I is surjective with operator norm at most 1. I is indicated in the bottom part of the diagram 1. Note that the bottom part of the diagram commutes.

Proposition 1.13. *For a potential current T ,*

$$\|T\|_{\mathbb{P}_d} = \sup_{\omega \in B_{\mathbb{P}^d}} \langle T, \omega \rangle. \quad (1.15)$$

It is not known whether also $\|\omega\|_{\mathbb{P}^d} = \sup_{T \in B_{\mathbb{P}_d}} \langle T, \omega \rangle$, cf. Conjecture 1.12.

1.4 Strong non-surjectivity

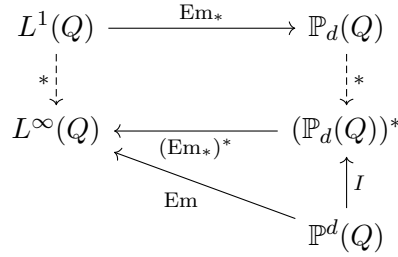


Figure 1: A diagram indicating the relationship between function spaces and potential spaces. The dashed arrows only indicate direction of duality. The full arrows correspond to operators. The bottom part of the diagram commutes.

We are in the following situation. We have three Banach spaces $Y = L^1$, $Z = \mathbb{P}_d$ and $X = \mathbb{P}^d$. Our goal is to show that $\text{Em}_*: Y \rightarrow Z$ is *not* surjective. By basic functional analysis, if $X = Z^*$, it would suffice to show that $\text{Em} = (\text{Em}_*)^*: X (= Z^*) \rightarrow Y^*$ is not surjective. Since $Y^* = L^\infty$, this is trivially the same as requiring that the PDE:

$$\sum_i f_i \det D\pi_i = \rho \quad \text{on } Q, \quad (1.16)$$

have a solution for every right hand side $\rho \in L^\infty$. Of course by a solution we mean that there exists a formal Lipschitz sum satisfying the equality a.e.

Since we are not able to prove that $X = Z^*$ (cf. Conjecture 1.12), we have to do some more work.

Definition 1.14. We say that the prescribed Jacobian equation has *strong Lipschitz non-regularity* if a sequence of $L^1(Q)$ functions u_n , $n \in \mathbb{N}$ can be found satisfying the two following inequalities simultaneously:

- (i) for each $n \in \mathbb{N}$, $\|u_n\|_{L^1(Q)} \geq 1$
- (ii) for each $n \in \mathbb{N}$,

$$\sup \left\{ \left| \int_Q u_n f \det D\pi \, d\mathcal{L}^d \right| : \max\{\text{Lip}(f), \|f\|_\infty\} \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\} \leq \frac{1}{n}.$$

Remark 1.15. Strong Lipschitz non-regularity is stronger than just “Lipschitz non-regularity”². In other words strong Lipschitz non-regularity implies that there exists some $\rho \in L^\infty$ for which the equation (1.16) does not have a solution. This is due to the open mapping theorem. If there existed a solution for every $\rho \in L^\infty$ then the operator $\text{Em}: \mathbb{P}^d \rightarrow L^\infty$ would be surjective. Then there would exist some open ball B in L^∞ containing 0 and wholly contained in

$$\left\{ \sum_i f_i \det D\pi_i : s\left(\sum_i (f_i, \pi_i)\right) \leq 1 \right\} = \overline{\text{conv}}\{f \det D\pi : s(f, \pi) \leq 1\}. \quad (1.17)$$

²This is called linearized Lipschitz non-regularity in [3]

Thus there would be some $\varepsilon > 0$ such that for every $u \in L^1$.

$$\sup \left\{ \left| \int_Q u f \det D\pi \, d\mathcal{L}^d \right| : \max\{\text{Lip}(f), \|f\|_\infty\} \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\} \\ \stackrel{\text{lin., conv.}}{=} \sup \left\{ \left| \int_Q u \rho \, d\mathcal{L}^d \right| : \rho \in \overline{\text{conv}}\{f \det D\pi : s(f, \pi) \leq 1\} \right\} \geq \varepsilon \|u\|_{L^1}. \quad (1.18)$$

This means either (i) or (ii) must fail for any sequence of u_n .

Theorem 1.16. *The prescribed Jacobian equation has strong Lipschitz non-regularity, if and only if the range of Em_* is not closed. In particular, if the equation has strong Lipschitz non-regularity, then the embedding Em_* is not surjective.*

Proof. By the open mapping theorem, the operator Em_* has closed range if and only if it is an isomorphism into $\mathbb{P}_d(Q)$. This is the case, if and only if there exists a constant C such that, whenever $u_n \in L^1(Q)$ satisfies (i) from Definition 1.14, then

$$\|\text{Em}_*(u_n)\|_{\mathbb{P}_d} \geq C. \quad (1.19)$$

Indeed, C corresponds to the operator norm of the inverse of Em_* . We calculate

$$\begin{aligned} \|\text{Em}_*(u_n)\|_{\mathbb{P}_d} &= \sup \left\{ \left| \text{Em}_*(u_n)(f, \pi^1, \dots, \pi^d) \right| : \|f\|_L \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\} \\ &= \sup \left\{ \left| \int_Q u_n f \det D\pi \, d\mathcal{L}^d \right| : \|f\|_L \prod_{j=1}^d \text{Lip}(\pi^j) \leq 1 \right\}. \end{aligned}$$

Therefore, we see that if a constant C satisfying (1.19) exists, then the sequence u_n does not satisfy (ii) of Definition 1.14. On the other hand, if a sequence u_n can be found such that both (i) and (ii) of Definition 1.14 hold, no such constant C can exist.

The final statement follows by Lemma 1.9. $\square$

Definition 1.17 (Strong non-surjectivity). Let X and Y be Banach spaces. We say that a linear operator $T: X \rightarrow Y^*$ is *strongly non-surjective* if for every $n \in \mathbb{N}$, there exists $\rho_n \in \frac{1}{n}B_{Y^*} \cap T(X)$ and a weak* open neighbourhood U_n of ρ_n such that

$$U_n \cap T(B_X) = \emptyset.$$

Theorem 1.18 (Predual separation theorem). *Let X and Y be Banach spaces and $T: X \rightarrow Y^*$ a bounded linear operator, which is additionally strongly non-surjective. Then there exists a sequence $v_n \in Y$ such that*

- (i) $\|v_n\|_Y \geq 1$,
- (ii) $\sup_{x \in B_X} |\langle T(x), v_n \rangle| \leq \frac{1}{n}$.

Proof. For details, see the paper. Here we will just indicate the basic idea. Firstly, we recall that the weak* continuous elements of Y^{**} are precisely those that coincide with the canonical images of elements of Y . In other words, any weak* continuous linear map $\varphi: Y^* \rightarrow \mathbb{R}$ admits some $y \in Y$ such that $\varphi(\eta) = \eta(y)$ for every $\eta \in Y^*$.

Next we use Hahn-Banach to strictly separate U_n from $T(B_X)$ with a weak* continuous linear functional, say u_n which is possible due to the definition of strong non-surjectivity. Now v_n is essentially obtained by normalizing u_n and looking at what “strict separation” really means. By what we wrote above v_n and u_n are both in Y . $\square$

Theorem 1.19. *Suppose $\text{Em}: \mathbb{P}^d \rightarrow L^\infty$ is strongly non-surjective. Then $\text{Em}_*: L^1 \rightarrow \mathbb{P}_d$ is non-surjective.*

Proof. Follows by the combination of Theorem 1.16 and Theorem 1.18 applied to $X = \mathbb{P}^d$ and $Y = L^1$. $\square$

2 Strong non-regularity of the Jacobian equation

1.todo

References

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